Group Topic: **Big Data Analytics for Student Dropout Prediction in Higher Education** Week No. 7

Git Repo/ Colab Link: https://github.com/prinxe0/Big-Data-Analytics-for-Student-Dropout-Prediction-in-Higher-Education/blob/main/notebook.py Date: April 23, 2026

Objectives

To understand and implement large-scale data procurement, gathering, and ethical ingestion procedures within a big data pipeline. The primary focus of this activity, though, is to fulfill the project requirements of acquiring a minimum dataset of 500,000 student records to make sure the data incorporates critical academic and psychological indicators while strictly adhering to the data anonymization and guidelines.

Tasks Completed

- Threshold Setup: Established collection parameters to meet or exceed the 500,000-row minimum pipeline requirement.
- Multi-Variable Sourcing: Procured a large-scale educational dataset containing 1,000,000 distinct student records.
- Feature Scope Validation: Verified that records embedded academic pressure metrics and multi-dimensional mental health indicators.
- Ethical Scrubbing: Checked and anonymized all data to ensure zero exposure of Personally Identifiable Information (PII).
- Distributed Ingestion: Loaded raw source text files into Spark Resilient Distributed Datasets (RDDs) using PySpark.

Tasks to be Completed

N/A

Results and Analysis

- Successfully gathered 1,000,000 distinct records, hitting exactly 200% of the project's 500,000-row minimum requirement.
- Spark's distributed architecture parallelized data loading smoothly, completing the ingestion of all 1,000,000 rows with zero memory failures.
- Compliance checks confirmed 100% anonymization. The collected rows maintained high feature density, successfully bundling critical risk parameters like burnout, anxiety, and sleep hours.

Challenges and Solutions

Challenge: Accumulating over 500,000 real student records threatens data privacy if personal identifying details enter the pipeline.

Solution: Enforced a strict ethical screening, permanently scrubbing all the PII at the sources so the final dataset contained only anonymized continuous features.

Conclusion

The lab has successfully met and exceeded the minimum requirement of the data-gathering objectives, securing a high-volume dataset of 1,000,000 rows. The activity proved that the collection of large datasets must integrate data ethics, comprehensive feature sourcing, and distributed processing from the outset. The robust dataset provides a secure, non-reactive foundation for subsequent predictive modeling phases.